Thermodynamic description of the Pb–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Pb–Sn binary system as calculated with pycalphad from the SGTE dataset PbSn-17Gie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Pb) + (Sn), lies at 182.7 °C.

1. Introduction

The Pb–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of W. Gierlotka, Y.-C. Tung, J. Phase Equilib. Diffus., 38, 814-28(2017). The calculations use pycalphad 0.11.2 at 1 atm.

Pure Pb melts at 327.5 °C. For Sn, the liquidus meets the axis at 231.9 °C.

2. Phase diagram

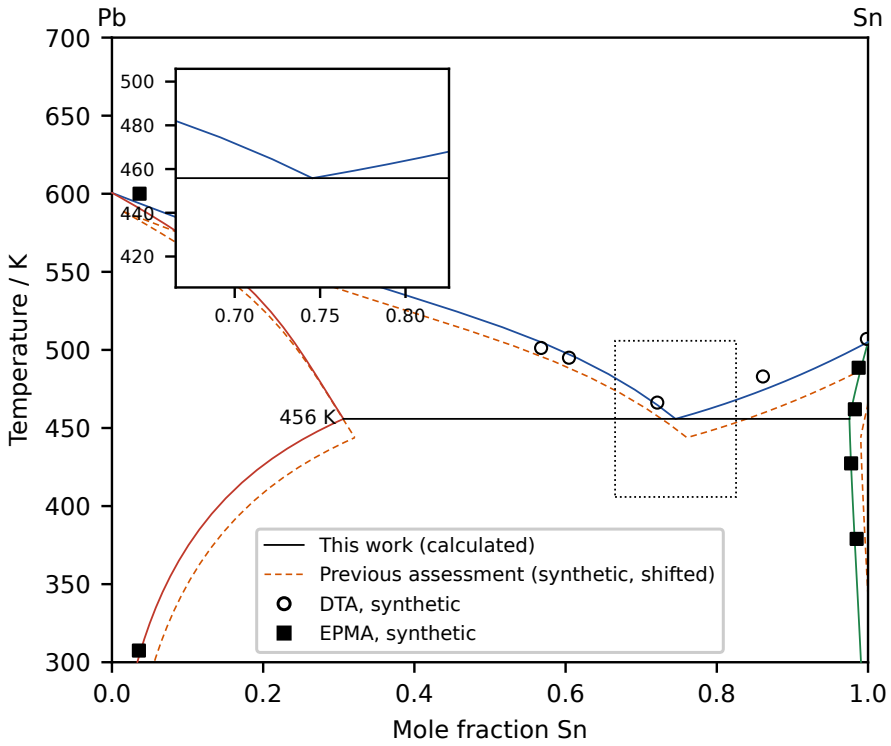


Fig. 1. Calculated Pb–Sn phase diagram with synthetic experimental-style points and a dashed, deliberately shifted second assessment (inset: enlarged view near the lowest invariant).

At 182.7 °C the eutectic L → (Pb) + (Sn) takes place; the L phase then contains 74.5 at.%.

Table 1. Invariant reactions in the Pb–Sn system (calculated).

Reaction	Type	T / K	Compositions (x(Sn))		
			phase 1	phase 2	phase 3
L → (Pb) + (Sn)	eutectic	455.8	0.745	0.306	0.975

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 700 K the integral mixing enthalpy of the liquid has its maximum, 1.29 kJ/mol, at 48.0 at.% Sn.

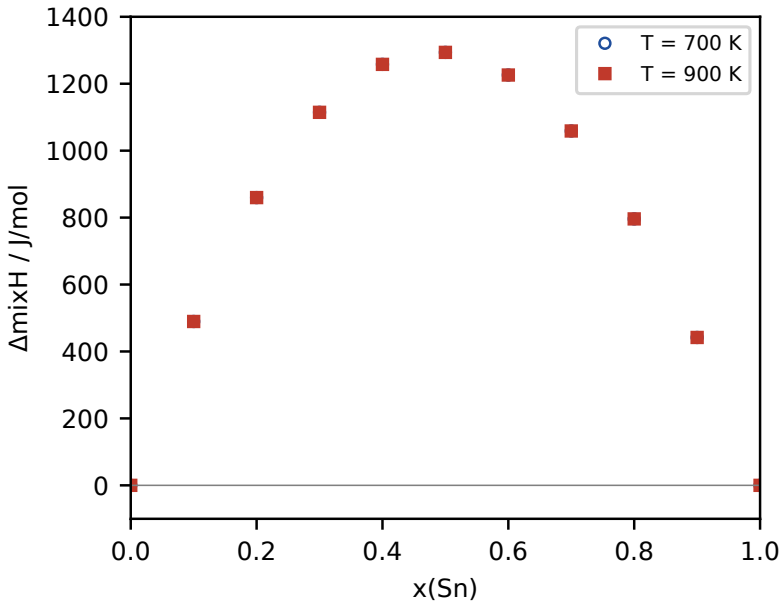


Fig. 2. Calculated mixing enthalpy of the liquid at 700 K and 900 K.

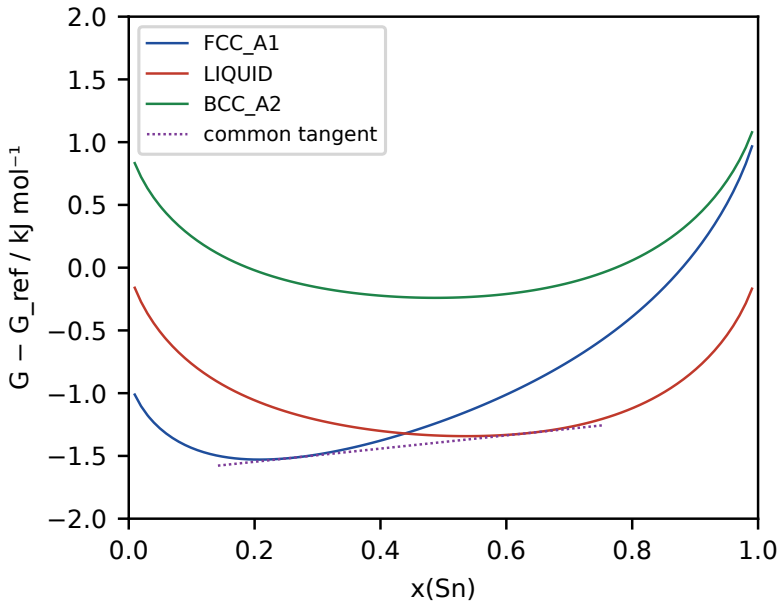


Fig. 3. Calculated Gibbs energy curves at 490 K.

Table 2. Redlich–Kister parameters of the liquid (Pb–Sn).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	5174.58	1.558	0.0
L1	331.69	0.0	0.0

Values as printed in the TDB file.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.